GLUCS: Gas Leak Detection and Safety System with Automated Response

Jayson Luseco
NU Dasmarinas
National University
Manila, Philippines
lusecojm@students.nu-dasma.edu.ph

Andre Sombillo
NU Dasmarinas
National University
Manila, Philippines
sombilload@students.nu-dasma.edu.ph

Jan Jerelle Sonio
NU Dasmarinas
National University
Manila, Philippines
soniojn@students.nu-dasma.edu.ph

John Paulo Patayon
NU Dasmarinas
National University
Manila, Philippines
patayonjs@students.nu-dasma.edu.ph

I. Introduction

The existence of gas leaks presents a major danger to residential and industrial settings because they cause fires and explosions while creating health risks. The absence of odor in natural gas allows the substance to gather unnoticed within confined areas thus creating conditions for disastrous events. Gas leak detection systems protect safety environments through their ability to notify people about dangerous gas exposure. Standard gas detectors warn users about gas presence but they do not actively prevent dangers from leaking gas.

The world has experienced numerous disastrous gas leak situations in recent years. The LG Polymers plant gas leak incident at Visakhapatnam India in 2020 caused 11 fatalities and led to more than 1000 individuals requiring hospital treatment because of toxic styrene gas emission (National Disaster Management Authority, India, 2020). Similarly, a gas explosion in Tondo, Manila, in 2019 left two people dead and several injured, emphasizing the urgent need for more advanced gas detection and response systems (Bureau of Fire Protection, 2019). The Bureau of Fire Protection (BFP) reports gas-related fires caused more than 2,000 incidents across the Philippines in 2020 thereby demonstrating the dangerous frequency of such gas accidents (BFP, 2020).

The proposed project supports three crucial Sustainable Development Goals set by the United Nations: SDG 3 (Good Health and Well-Being), SDG 9 (Industry, Innovation, and Infrastructure) as well as SDG 11 (Sustainable Cities and Communities). Our system extends safety protection through an integration of ESP32 microcontroller with Blynk IoT platform which enables both immediate monitoring and automated responses for remote alerts. Through an enhanced process, our system distinguishes itself from conventional detectors by performing leak detection operations as well as activating ventilation systems and delivering instant voice alerts to users. When the system detects leaks the servo-controlled gas valve together with flame sensors immediately cuts off gas supply to prevent fire hazards.

Voice alarms serve as crucial elements for emergency response measures particularly when industries demand fast evacuation or corrective actions. Our system provides voice alerts which deliver precise instructions to users thus lowering their panic reactions and boosting their response rate. Microcontrollers working with automated response mechanisms bring advanced gas leak safety technology because they detect issues before they happen and automatically activate mitigation systems.

Currently available gas detection systems produce important alerts yet they do not automatically execute interventions to reduce potential damage thus preventing catastrophic situations. Our proposed system unites smart sensors and real-time monitoring and automation which provides complete safety protection for residential buildings as well as industrial facilities and business establishments. The implementation of IoT and automation allows this project to advance safer and smarter sustainable environments.

II. Review of Related Studies

Both residential areas and industrial facilities face significant safety threats when gas leaks occur, as they can lead to fatal fires, explosions, and the release of toxic gases. Gas-related accidents in the Philippines contribute to serious safety concerns, with thousands of such incidents reported annually. Traditional gas leak prevention systems typically rely on basic gas sensors that activate warnings when unsafe gas concentrations are detected. However, these systems often prove inadequate, as they only address detection and lack comprehensive solutions that protect against both short-term and long-term dangers. Recent advancements in technology have led to the development of more sophisticated systems that not only detect gas leaks but also incorporate automated safety measures. These systems enhance safety by activating responses such as ventilation and gas flow control, effectively reducing the risk of disasters [1].

Modern gas leak detection systems achieve two major objectives by detecting leaks as well as implementing proactive measures to minimize dangers. The implementation of automated response features within gas detection platforms reduces the extent of both human casualties and property destruction from gas leakage events. An automated system employing ventilation activation triggered by gas detection allows the prevention of confined space gas accumulation thus avoiding potential fire or explosion incidents[3]. The automated systems minimize human involvement at critical periods since immediate actions are essential while manual reaction delays might produce serious outcomes.

Traditional gas leak detection systems fail to address fire and explosion dangers during a gas leak because they issue alerts only after leak detection occurs. The project we built eliminates these weaknesses through its dual approach of real-time gas leak detection with integrated immediate safety automation systems. The system enables automatic operation of safety measures which include gas valve cutoff and exhaust fan activation in case of gas leakage. The automated safety features of this system stop hazardous gas accumulation because they significantly reduce the chance of both fire and explosion occurrences [13]. Both detection systems and automated safety protocols form a necessary combination which enhances the protection within residential and industrial settings.

The project design includes a voice alarm system that becomes crucial for environments dealing with high-risk situations. Traditional gas leak detection devices notify the presence of gas leakage to people but they fail to offer complete directions regarding proper response actions. A voice alarm system provides detailed step-by-step instructions to follow during a gas leak to guide people either toward evacuation or toward turning off gas supply. The research shows voice alarms surpass standard alarms as an industrial safety measure because they provide better response efficiency in environments that lead to accidents when responses are delayed [1].

IoT technology enables this project to track and monitor data in real time along with providing remote capabilities. The system provides special benefits to people who need to monitor their equipment from different locations. Through its mobile application users gain remote access to gas levels along with safety system status which enables swift action from users who are absent from the site [14]. The remote monitoring feature demonstrates its worth by allowing users to take prompt actions when needed thus it improves safety and minimizes response times and accidents.

Through its cost-effective solution the project supports Philippine gas leak risk reduction strategies by providing an implementable technology that works in residential and industrial environments. A Bureau of Fire Protection [2] report reveals gas leakages as the main cause of fires affecting the country. The yearly statistics for 2020 revealed a total of 2000 gas-related fires in the country. The complete gas leak remediation solution offered by automatic systems which activate ventilation systems combined with gas supply interruptions exceeds traditional detection system capabilities.

The project minimizes gas-related fires as one of its functions while simultaneously resolving household gas leak problems. The National Power Corporation [5] analyzed data which showed that more than 10% of Philippine house owners encountered gas leaks somewhere in their residence. Through remote gas level monitoring the system enables homeowners to detect emerging problems that potentially threaten lives and property before such issues become hazardous.

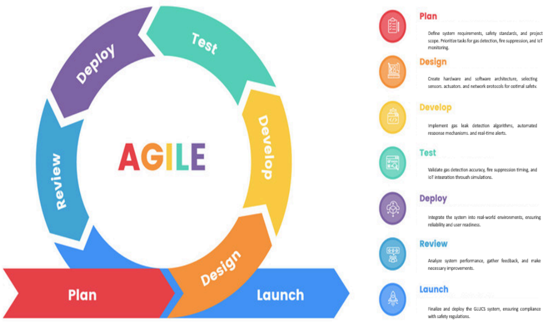
Real-time monitoring, voice alarms and automated security features form a complete solution to Philippine gas leak issues as part of this project. This system does not only identify leaks in the environment but it actively works to prevent explosions and fires resulting in substantial protection of residential, commercial and industrial areas.

III. Methodology

The GLUCS project development implements Agile methodology through iterative development which performs features via short testing cycles of 2-4 weeks each. The GLUCS project adopts Scrum as its framework with the Product Owner defining system requirements while the Scrum Master facilitates Agile processes and the Development Team handles engineering and testing. The project tracks development progress through daily status check meetings that also resolve issues along with sprint planning sessions which prioritize distinct sets of tasks for each development cycle. Once each sprint is complete the team will present finished features in a review then use the retrospective to find improvement opportunities.

The Product Backlog organizes development by collecting all essential system requirements including gas leak detection together with valve control and fire suppression and IoT-based monitoring. The sprint process will handle multiple essential tasks from the top priority list to achieve continuous development. The system will retain reliability through constant integration alongside testing which includes automatic checks for gas detection precision alongside real-fire and gas leak simulations. The system performance evaluation at various stages helps to optimize the response functions.

The detection algorithms and response efficiency and user experience will be improved through user feedback that is implemented after each sprint cycle. Through continuous iterations GLUCS develops into a secure automated gas leak detection system that fulfills top safety standards by adapting to genuine world needs.



A. Plan

Firstly, to implement the system, the team will engage in research to identify the suitable gas sensors that need to be installed in this case, it is the MQ-2 gas sensors and an esp32 microcontroller, which forms the central component of the system. They, however, will also identify and select the right hardware components, such as the exhaust fans, DFPlayer mini, speakers, fire sensor, MG995 servo, and water pump in consideration of compatibility with the esp32 system. After the selected components are chosen, the team will go about the process of connecting the hardware to the microcontroller. In this stage, they shall individually run all the components to verify their operational status, as the exhaust fan and voice alarm, for instance, will trigger on gas and fire. After the completion of the hardware configuration, the team will go for the software part, where the esp32 microcontroller will be programmed to analyze sensor data and make decisions based on the gas concentration, smoke level, and water level.. The program will include the gas level that triggers the exhaust fan and the alarm voice alarm, and fire detection with an extinguishing system beyond which the dangerous gas accumulates and fire has been detected. Following coding, the team is then to conduct a validation whereby the team will induce gas leaks and even adjust the calibration of the other sensors to test the authenticity of the readings taken as well as respond to the system accordingly. At this latter stage, the team will put in place necessary safety measures and fine-tune the system to its response to such gas leaks, the right 'sound,' the voice alarm, the valve with the integration of the servo, and the fire detection with the extinguishing system. Last but not least, the final assembling into the diorama and system document will be written, which will describe the design of the hardware, the software code implemented, and the testing of the system, and results obtained will be written in the report by the team. This approach makes it possible for the creation of a credible gas leak detection system coupled with automated responses in residential as well as industrial applications.

B. Design

1.) Block Diagram

The block diagram consists of four sections which include gas detection system, smoke detection system, fire detection system and app user interface.

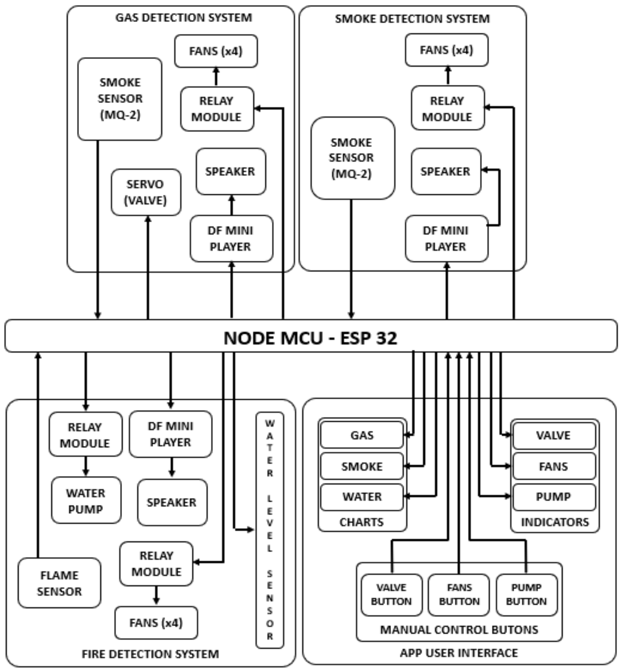


Fig. 1 - System Block Diagram

Fig. 1 depicts an automatic safety system that employs an ESP32 NodeMCU with features for fire detection along with smoke detection and gas detection which integrates Blynk IoT for distant system control functions. The system includes multiple sensors and actuators that use mobile interface control for hazard detection which activates predefined response sequences.

An MQ-2 sensor activates four fans and a servo valve and triggers an alarm through a speaker and DF Mini Player when it detects gas. Another MQ-2 sensor functions as a smoke detector that activates user alerts through the Blynk IoT dashboard similarly to how it operates.

The implementation of fire detection depends on a flame sensor coupled with a water level sensor. When fire detection occurs the ESP32 simultaneously activates both fans along with the pump system while triggering the alarm mechanism. Water level sensor implements a live feed of fire protection supply status on the Blynk platform.

Users can access real-time updates on gas and smoke as well as fire and water levels through Blynk IoT platform software to operate fans together with pumps and valves remotely. The combination of ESP32 with Blynk IoT helps boost fire protection while rapidly notifying system users through automatic safety operations and current device status reporting.

2.) Flowchart

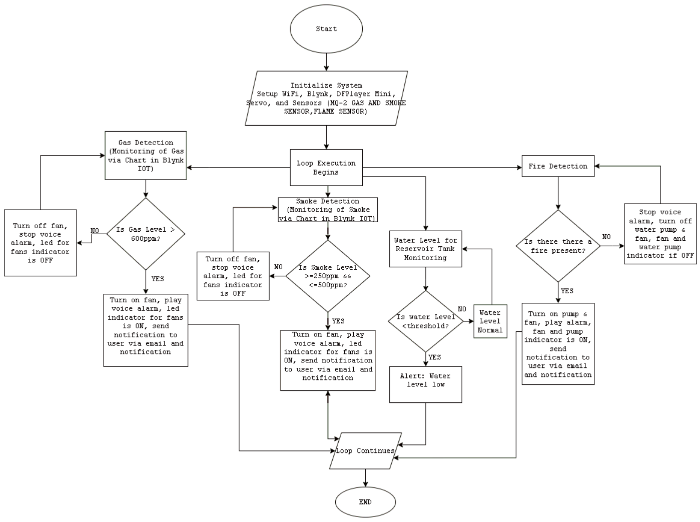


Fig. 2 - Flowchart

Fig. 2 demonstrates sensor operations for identifying safety hazards followed by data analysis to automatically activate safety components which include fans, alarms, a water pump and a gas shut-off valve for efficient hazard response and mitigation.

The GLUCS system starts its functions after turning on by powering up WiFi together with Blynk connection and DFPlayer Mini audio alerts plus servo motor control and sensor and actuator pin definition. After the successful initialization the system maintains continuous operational loop status to track environmental conditions.

The gas sensor determines gas concentration first. The system determines no leak exists when the gas levels measure below 600 by enabling fan and alarm systems. When the gas sensor readings pass 600 ppm the device activates ventilation through the fan while playing an alarm with the DFPlayer Mini and records the incident in the Blynk dashboard.

After that the system performs a flame detection utilizing a flame sensor. The system stops all operations by disabling the alarm and turning off both the water pump along with the fans when no flame signal is present. The system starts water pumping and activates fans along with alarm operation when it detects a flame because it acknowledges a fire emergency. The event also logs into Blynk for remote monitoring. The water level sensor controls the monitoring process for the water reservoir at the same time. The system will display "Water Level is Normal" in Blynk when the water reaches or exceeds its specified threshold. The system creates a “Water Level is Low” warning message through which it notifies users about dropping water levels when thresholds are crossed.

The smoke sensor calculates the degree of smoke density within the protected area. The system maintains operation with no alarms and keeps the fan disabled at all times when smoke levels remain below 100. When the smoke measurement falls within 250 to 500 the system turns on ventilation fans and starts an alarm which creates Blynk system records of the event.

System users can access manual controls through Blynk to operate the servo motor for ventilation access and control the exhaust fan and water pump activation status.

The system operates in a repetitive cycle for continuous time-based monitoring followed by instant responses when detecting gas leaks alongside fire events or smoke occurrences or insufficient water in the reservoir to improve safety alongside environmental consciousness.

3.) Blynk IoT - Web Dashboard, Datastreams, Events and Notifications

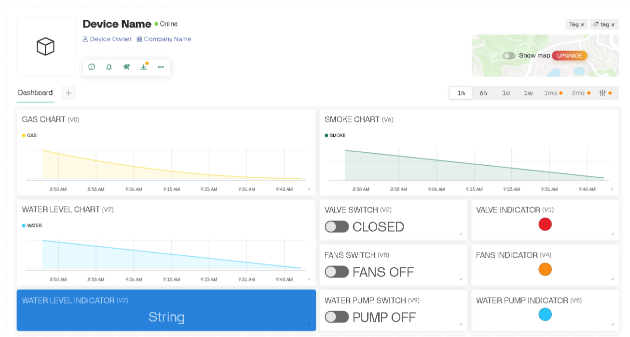
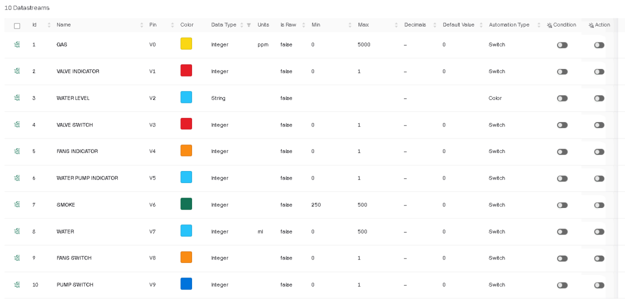


Fig. 3A - Web Dashboard

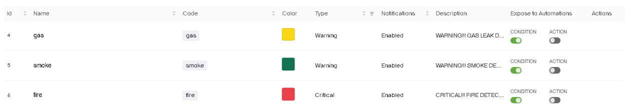
Fig. 3A shows the dashboard operation of sensors while managing environmental measurements in addition to actuating devices. The system displays actual time graphical representations of gas measurements alongside smoke observations and water amount displays. The system switches allow users to operate the valve and control the fans and water pump function through visual indicators that display the active devices. The water level indicator gives continuous updates about the current situation. The system provides an efficient approach to monitor air quality alongside the management of fires and control water levels.


10 Datastreams

Id	Name	Pin	Color	Data Type	Units	Is Raw	Min	Max	Decimals	Default Value	Automation Type	Condition	Action
1	GAS	V0		Integer	ppm	false	0	5000	–	0	Switch		
2	VALVE INDICATOR	V1		Integer		false	0	1	–	0	Switch		
3	WATER LEVEL	V2		String		false			–		Color		
4	VALVE SWITCH	V3		Integer		false	0	1	–	0	Switch		
5	FANS INDICATOR	V4		Integer		false	0	1	–	0	Switch		
6	WATER PUMP INDICATOR	V5		Integer		false	0	1	–	0	Switch		
7	SMOKE	V6		Integer		false	250	500	–	0	Switch		
8	WATER	V7		Integer	ml	false	0	500	–	0	Switch		
9	FANS SWITCH	V8		Integer		false	0	1	–	0	Switch		
10	PUMP SWITCH	V9		Integer		false	0	1	–	0	Switch		

Fig. 3B - Datastreams

In Fig. 3B, the system sensors and controls are documented in this table which includes their identification numbers along with names and data types and specified functions. The system contains gas detection devices as well as smoke and water level sensors to monitor the environment in addition to valve switches and fan and water pump indicators. The system enables time-sensitive observation along with automatic environmental control functions.



Id	Name	Code	Color	Type	Notifications	Description	Expose to Automations	Actions
4	gas	gas		Warning	Enabled	WARNING!!! GAS LEAK D...	CONDITION ACTION	
5	smoke	smoke		Warning	Enabled	WARNING!!! SMOKE DE...	CONDITION ACTION	
6	fire	fire		Critical	Enabled	CRITICAL!!! FIRE DETEC...	CONDITION ACTION	

Fig. 3C - Events and Notifications

Fig. 3C presents warning and notification configurations to detect gas and smoke and fire. The table contains entries that use IDs and names and display color alerts (yellow for gas and green and red for smoke and fire) to indicate notification states. Only gas and smoke detection warnings are turned on because fire detection has critical status. The active automation settings guarantee that alerts automatically activate required responses.

C. **Develop**

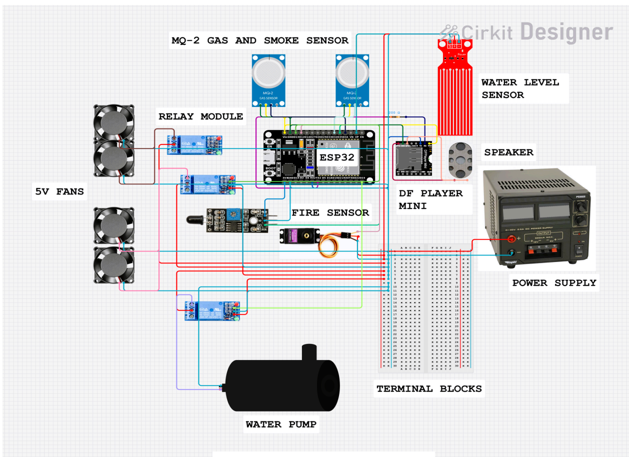


Fig. 4 - Circuit Diagram

Fig. 4 shows an ESP32 microcontroller interconnected with gas and smoke sensors, water level detectors, fire alarms along with actuators and other modules. The ESP32 receives air quality data from the MQ-2 gas and smoke sensor during its wiring. The setup includes a water level sensor which functions as a water detection device. The circuit includes a fire sensor that triggers signals to reach the ESP32 in case of fire detection.

Furthermore the DFPlayer Mini module connects to a speaker to produce audio output while being linked to the ESP32. The relay module directs simultaneous operations of multiple devices. A relay module operates 5V ventilating fans through its management system. The same relay activates a water pump for required operations.

Portal blocks maintain efficient power distribution as a regulated power supply transfers voltage into the circuit system. A precise arrangement of wires provides both communication pathways and power delivery routes for all components. The system checks for gas and smoke and fire incidents by using its programmed responses to initiate fan operation and alarm activation and water pump function.

IV. Testing and Develop

1. System Performance Evaluation

This section presents the results of the gas leak safety system's performance based on various test scenarios. The evaluation includes gas and smoke detection accuracy, response time, and the effectiveness of the exhaust and extinguishing mechanisms.

1.1 **Gas and Smoke Detection**

The MQ2 sensor was tested with different concentrations of gas (LPG, smoke, and butane). The detection threshold and sensor accuracy were analyzed.

Gas Type	Valve	Concentration (ppm)	Detection Time (s)
LPG	YES	600 - 4095	1
Smoke	NO	250 – 500	1
Butane	YES	600 - 4095	1

Expected Output	Testing	Actual Output
Fan ON, Valve ON (Smoke) ppm (600 – 4095)	Test I	Fan ON, Valve ON (Smoke) ppm (600 – 4095)
	Test II	Fan ON, Valve ON (Smoke) ppm (600 – 4095)
	Test III	Fan ON, Valve ON (Smoke) ppm (600 – 4095)

1.2 **Exhaust System Efficiency**

The exhaust fans were activated upon detecting hazardous gas levels. The efficiency of the exhaust system in reducing gas concentration was measured over time.

Time (s)	Initial Gas Level (ppm)	Gas Level After Exhaust (ppm)
0	0	0
10	4095	<100
20	4095	<100

Expected Output	Testing	Actual Output
Gas and Smoke Level (600 - 4095), Fire Detected	Test I	Gas and Smoke Level (600 - 4095), Fire Detected
	Test II	Gas and Smoke Level (600 - 4095), Fire Detected
	Test III	Gas and Smoke Level (600 - 4095), Fire Detected

2. Fire Detection and Extinguishing System Performance

The fire sensor was tested under controlled fire conditions, and the response time for the activation of the water pump was recorded.

Fire Source	Activation Time (s)	Water Pump Duration (s)
Lighter	1	5

Expected Output	Testing	Actual Output
Water Pump ON, Fans ON	Test I	Water Pump ON, Fans ON
	Test II	Water Pump ON, Fans ON
	Test III	Water Pump ON, Fans ON

The water level sensor ensured a consistent water supply in the reservoir tank. The minimum water level required for efficient operation was determined.

Water Level (ml)	Status
250-500	Normal
0-250	Low

Expected Output	Testing	Actual Output
0-250 Low 250-500 High	Test I	0-250 Low , 250-500 High
	Test II	0-250 Low, 250-500 High
	Test III	0-250 Low, 250-500 High

3. Voice Alarm and IoT-Based Monitoring

The DFPlayer Mini and speaker were tested for clear voice alerts upon detection of gas, smoke, or fire. The effectiveness of the Blynk app in providing real-time notifications and manual control was evaluated.

Event	Voice Alarm Triggered	Blynk Notification Received	Response Time (s)
Gas Detected	YES	YES	1
Smoke Detected	YES	YES	1
Fire Detected	YES	YES	1

Expected Output	Testing	Actual Output
Gas Detected, Smoke Detected, Fire Detected	Test I	All working
	Test II	All working
	Test III	All working

The results indicate that the MQ2 sensor successfully detects gas leaks with high accuracy, and the exhaust system effectively reduces gas concentration. The fire sensor promptly detects fire, and the automated extinguishing system responds efficiently. Additionally, the integration of a voice alarm and Blynk app enhances real-time awareness and control, providing immediate alerts and allowing manual intervention when necessary.

Overall, the system demonstrates significant potential for improving safety in residential and industrial environments by ensuring rapid detection and response to gas leaks and fire hazards.

V. Conclusion

The real-time detection, automated safety protocols, and remote monitoring capabilities of the GLUCS: Gas Leak Detection and Safety System with Automated Response were successfully addressed by the key limitations found in conventional method gas leak detectors. However, the system has successfully proved to be very effective at detecting gas leakages, triggering necessary safety measures, and sending real time alerts to users. The system achieves overall safety advancement for residential and industrial areas through a synergy of gas sensors, micro control mechanism, automated control mechanism and IoT based applications.

During testing, the system responds within seconds to hazardous gas leaks and almost instantaneously takes preventive measures. This allows the gas to be exhausted though an exhaust fan, setting off a voice alarm and turning off the gas valve, significantly reducing the chances of gas accumulation and explosion. In addition to that, the fire detection and suppression system provides one more layer of safety which offers swift actions against any fire hazard. With the addition of IoT technology, the system makes the users' usage easy, they can monitor in real time, control with a given distance and receive notifications instantaneously to guide users know when they can respond.

Among the most outstanding features of the GLUCS system is its ability for automatically responding, and thus makes GLUCS hardware different from standard gas detection instruments. Compared with conventional systems, which usually alert users to a possible leak only with alarms, the GLUCS system actively curtails risks through the production of corrective procedures, including shutting down gas supply and triggering fire suppression. This type of proactive approach greatly reduces accident potential and thus makes the system highly innovative in safety technology.

The system is effective, but could still be improved in some respects. Artificial intelligence (AI) algorithms could also be included to predict and analyze gas leak patterns in the future and would make the hazard detection even more faster and precise. Further, the system could prove useful when such a range of gas sensors can be used in the industrial environments where different types of hazardous gases exist. An additional improvement could be if the system were to be operated with solar power, rendering it more sustainable and less likely to fail in electricity outages.

Overall, the GLUCS system greatly improves gas leak safety with the features of accidental gas detection, immediate faulty situation emergency response, fire suppression, and distance monitoring. This aligns with United Nation Sustainable Development Goals such as SDG 3 (Good health and well being), SDG 9 (Industry, innovation and infrastructure) and SDG 11 (Sustainable cities and communities) because of the promotion of a safer living and working environment. Modern technology is used to leverage the GLUCS system to set the standard in gas leak detection and safety so that people and properties stay safe from potential risks. This project represents a base of development for the further creation of smart safety systems in the residential, commercial and industrial settings.

As such both safety and technological upgrade continues GLUCS will be one of the roles systems like this will play in creating safer environments, away from potential hazards of leaks and fire. This project succeeded to show the value of innovation, automation, connectivity in safety solutions, leading to a safer, more resilient future.

References

[1] Santos, J. D. (2020). *Advancements in gas leak detection systems in the Philippines: A focus on automation and safety measures*. Philippine Journal of Industrial Safety, 17(4), 125-134.

[2] Bureau of Fire Protection (BFP). (2020). *Annual report on fire-related incidents and gas leaks in the Philippines*.

[3] Santos, E. L., & Mendoza, C. P. (2020). *Integrating flame sensors with gas detection and suppression systems for improved fire safety*. Philippine Fire Safety Review, 24(2), 101-110.

[4] Alam, M. A., & et al. (2018). "Arduino-Based Safety Systems: Applications and Benefits." *Journal of Engineering and Technology*, 45(3), 155-165.

[5] National Power Corporation (NPC). (2021). *Gas leak statistics and safety protocols in residential areas*.

[6] United Nations. (2015). *Transforming our world: The 2030 Agenda for Sustainable Development*. UN Publishing.

[7] Lee, J., & Park, H. (2020). "Fire Suppression in Gas Leak Detection Systems." *Journal of Fire Protection Engineering*, 29(2), 67-79.

[8] Rivera, M., Torres, G., & Ramirez, S. (2022). "Integrated Safety Systems: A Holistic Approach to Hazard Detection." *Journal of Safety Engineering*, 37(4), 102-117.

[9] Jolhe, D., Ingle, A., & Gajanan, D. (2013). Efficient gas leakage detection and control system using GSM module. *International Journal of Engineering Research and Technology (IJERT)*.

[10] Kumar, S., Gupta, R., & Singh, A. (2020). GSM-based gas leakage detection and alerting system. *International Journal of Scientific and Engineering Research (IJSER)*.

[11] Sharma, P., Kaur, R., & Singh, M. (2022). Smart fire suppression systems: Integration of water level sensors and automated alerts. *Journal of Safety Engineering*, 11(2), 45–52.

[12] Sodiq, J. O. (2024). Gas leakage detector and monitoring system. *ResearchGate*.

[13] De Guzman, M. S. (2021). Development of a gas leak detection system using ESP32 microcontroller. *Philippine Journal of Technology*, 8(1), 52-60.

[14] Gupta, S., & Ramesh, T. (2020). *Remote monitoring using IoT in safety applications*. International Journal of Emerging Technologies, 8(2), 98-112.

[15] Singh, A., & Rajan, S. (2018). *Smart fire detection and response systems: An integrated approach*. Fire Prevention and Control Journal, 12(3), 34-49.

[16] Goyal, R., Sharma, P., & Verma, K. (2021). *Advancements in fire suppression: A review on automated extinguishing systems*. Fire Safety Journal, 45(2), 210-225.

[17] Bureau of Fire Protection (BFP). (2020). *Annual report on fire-related incidents and gas leaks in the Philippines*.

[18] National Disaster Management Authority, India. (2020). *Report on the LG Polymers gas leak disaster*.

[19] Bureau of Fire Protection (BFP). (2019). *Gas explosion incident in Tondo, Manila*.